

POINT IROQUOIS, MI, USA

WMO: 997265

Lat: 46.4846N

Lon: 84.6309W

Elev: 611

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.7	-4.1	N/A	N/A	N/A	N/A	N/A	N/A	47.3	21.5	38.1	24.0	8.5	120	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	81.8	N/A	78.1	N/A	74.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 34.6	28.6	24.7		-16.1	86.6	7.9	2.8	-21.8	88.7	-26.4	90.3	-30.9	91.9	-36.6	94.0	
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	16.2	16.3	24.3	36.9	48.2	58.3	64.6	64.8	58.4	47.3	35.6	24.7
	DBStd	19.19	11.22	10.44	10.49	7.19	7.93	6.82	5.91	4.74	6.53	7.30	8.15	9.08
	HDD50	4693	1047	944	796	399	130	9	0	0	7	142	436	783
	HDD65	8765	1512	1364	1260	844	526	220	81	61	216	550	883	1248
	CDD50	1567	0	0	1	4	74	258	452	459	258	58	3	0
	CDD65	165	0	0	0	0	5	19	68	55	17	1	0	0
	CDH74	831	0	0	0	1	61	166	328	201	68	6	0	0
	CDH80	164	0	0	0	0	17	39	60	36	12	0	0	0
	WSAvg	10.0	11.1	10.4	11.1	11.5	10.1	8.3	7.4	8.3	8.8	10.4	10.8	12.4
	PrecAvg	34.4	2.3	1.6	2.0	2.5	2.7	3.0	3.1	3.0	4.2	3.8	3.4	2.7
(15) Precipitation	PrecMax	45.7	4.4	3.3	4.8	4.9	5.1	6.2	6.7	4.6	8.9	6.8	6.9	5.1
	PrecMin	24.4	0.9	0.2	0.2	1.1	0.7	0.8	1.1	1.4	0.7	1.1	1.3	0.2
	PrecStd	5.6	0.9	0.8	1.1	1.1	1.1	1.1	1.4	0.9	1.9	1.6	1.4	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	42.5	54.0	65.6	82.5	85.6	85.6	84.3	81.5	72.8	60.3	44.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	37.7	35.2	46.8	56.1	72.1	79.1	81.6	79.7	75.0	66.2	52.6	41.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	35.3	32.9	41.3	50.4	65.2	73.4	77.9	75.7	71.0	61.5	48.7	38.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	32.5	30.8	37.3	45.7	59.8	68.5	74.1	72.2	67.8	57.6	46.0	36.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.5	14.4	11.5	14.5	14.2	13.3	11.4	12.8	10.1	9.5	10.0
		MCDBR	10.5	13.3	17.6	21.2	25.8	23.2	20.9	18.2	17.9	15.6	13.0	10.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.281	0.291	0.319	0.355	0.371	0.389	0.390	0.378	0.367	0.332	0.303	0.286
	taud		2.344	2.302	2.334	2.332	2.374	2.346	2.365	2.399	2.424	2.503	2.509	2.382
	Ebn at Noon		259	280	288	285	284	278	276	275	268	263	252	243
	Edh at Noon		23	30	34	37	37	38	37	35	31	25	20	20
(44) All-Sky Solar Radiation	RadAvg		340	632	1030	1411	1763	1901	1912	1617	1206	701	366	264
	RadStd		36	68	118	161	142	128	108	106	90	81	48	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page